WSABI Summer Research Lab **Summer 2026**

Lab 7: Causal Inference

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7.1 NBA Timeouts After a 7-Point Drop

In Lecture 7, we used NBA timeouts after a 7-point drop in score margin over the last 3 minutes as a running example for causal inference. In this lab, you will build that analysis yourself from NBA play-by-play data downloaded with `hoopR`.

To keep the estimand consistent, use the following operationalization throughout the lab:

- Use regular-season NBA play-by-play from the `2021-2022` through `2025-2026` seasons. In `hoopR`, these correspond to `seasons = 2022:2026`.
- For each team, track the change in score margin over the previous 180 seconds of game time.
- Define a **decision opportunity** as the first opponent scoring play that pushes that 180-second score-margin change to `-7` or worse.
- Define treatment as whether the coach of the team facing that swing calls a timeout immediately, before the next clock change.
- Require that there was no timeout in the previous 180 seconds of game time.
- After recording a decision opportunity, do not count another opportunity for that same team for the next 180 game-seconds. An actual timeout also ends the current episode.
- Define the outcome as the change in score margin over the next 180 seconds of game time.

This is only one of several reasonable ways to define the timeout problem, but it gives everyone the same unit, treatment, outcome window, eligibility rule, and episode rule. That makes your analyses easier to compare across methods and across students.

7.1.1 Data

You will download play-by-play data with `hoopR::load_nba_pbp()`. After the starter code constructs the timeout-opportunity dataset, the main observed pre-treatment covariates available for adjustment are:

- `swing_last_180`: the change in score margin over the previous 180 seconds;
- `current_margin`: the team's score margin at treatment time;
- `period_number`: the game period;
- `end_game_seconds_remaining`: game time remaining;
- `home_focal`: whether the team at risk is the home team;

- `pregame_spread_focal`: a pregame team-strength proxy derived from the home-team spread;
- `spread_missing`: an indicator that the spread value was unavailable.

The starter code also keeps identifiers such as `season`, `game_id`, and `focal_team_id`. By contrast, many substantively relevant covariates are *not* observed in this lab dataset, including lineup quality, fatigue, private tactical information, and detailed matchup context.

7.1.2 Your Task

Create one row per decision opportunity using the operationalization above. Your final analysis dataset should include at least:

- a game identifier and season;
- the 180-second score-margin swing at time zero;
- the current score margin at time zero;
- the period and game time remaining;
- a home/away indicator for the team at risk;
- a team-strength proxy such as pregame spread, if available;
- the treatment indicator `timeout_now`;
- the outcome `margin_change_next_180`.

Then do all of the following:

1. Report how many decision opportunities are in your final dataset.
2. Report the fraction of those opportunities where the coach called timeout immediately.
3. Compute the naive difference in mean outcomes between timeout and no-timeout moments.
4. In words, state the unit, treatment, control, outcome, and target estimand.
5. List at least five pre-treatment covariates available in this dataset that you plan to adjust for, and give one example of a bad control.

7.2 Propensity Model and Overlap

In the lecture, we emphasized that observational causal inference tries to reconstruct the controlled comparison we wish we had. A key step is modeling treatment assignment and checking whether timeout and no-timeout moments are actually comparable in the observed data.

7.2.1 Your Task

Using logistic regression, model the probability that the coach calls timeout immediately. You should use only **pre-treatment** covariates. At a minimum, your propensity model should include:

- `swing_last_180`;
- `current_margin`;
- `period_number`;
- `end_game_seconds_remaining`;
- `home_focal`;
- `pregame_spread_focal`;
- `spread_missing`.

You may include additional observed pre-treatment variables such as season or team indicators if you can justify them, but do not include post-treatment variables or outcomes.

Then:

1. Plot the estimated propensity score distributions for timeout and no-timeout moments.
2. Briefly describe whether overlap looks strong or weak.
3. Use the starter-code balance function to compare covariate balance before adjustment.

7.3 Estimating the Effect

You will now estimate the timeout effect in several different ways. The starter code provides scaffolding for each estimator. Your job is to plug in the propensity model and interpret the results.

7.3.1 Your Task

Estimate the timeout effect using all of the following:

1. a naive difference in means;
2. regression adjustment;
3. nearest-neighbor matching on the propensity score;
4. inverse-probability weighting (IPW).

For this part:

1. Build a short table that reports the estimate from each method.

2. Check post-adjustment balance for your matching and weighting analyses.
3. State whether the methods broadly agree or disagree.
4. Give one plausible reason why the estimates might differ.

7.4 Sensitivity and Interpretation

Even after adjustment, the timeout analysis may still suffer from hidden confounding. Coaches see many things that are not stored in the public play-by-play data.

7.4.1 Your Task

Use the provided sensitivity-analysis code on your regression-adjusted model. Then answer all of the following:

1. What kinds of unmeasured confounding are still plausible in this setting?
2. How sensitive do your conclusions appear to hidden bias?
3. What is the strongest causal claim you are willing to make from this analysis?
4. What would make the design more convincing?